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Chapter 3

Forecasting Overview



Introduction

Forecasting provides a method of avoiding costly stockouts and inventory excesses which inhibit cash flow. This is particularly true when the forecasts are used to drive the creation of supply plans within DRP and MPS/MRP modules.

The Forecasting module contains seven simple forecasting methods. They are methods which people use every day, such as:

- Whatever was sold in the last three periods divided by three is what will be sold in the next period.
- Whatever was sold in the last twelve or thirteen periods divided by twelve or thirteen is what will be sold in the next period.
- Whatever was sold in the corresponding next three periods of last year divided by three is what will be sold in the next period.

These are simple forecasting methods. Forecasting selects the one best method to forecast this item, or group of items, at this moment in time. This is done through a process of simulation, where the module goes back in time, evaluating each formula for each item or group of items.

The module projects these simulated periods, using each of the methods, pretending they did not happen. Whichever forecast method best projects these periods that have already happened, by comparing their forecasts with actual data, is the one chosen to predict the future.

Forecasting goes through this simulation every time you run the period-end forecast. It is a dynamic simulation. To pick the one forecast method that worked best, the module calculates a forecast error percentage. The closer to zero the error, the better the method. The actual sales are extracted from the Sales Analysis module.

Relationship to Other Modules

Forecasting is a fully standardised module, and as such operates under the control of System Manager. Forecasting is one of the Group of Advanced Modules which extend the functionality of the Base Modules.

The modules which are pre-requisite for the operation of Forecasting are:

- Inventory Management.
- General Ledger.
- System Manager.

Item information is extracted from the Inventory module. Sales History can be extracted for these items from the Sales Analysis module.

However, if these modules are not installed, it may be possible to obtain this information from equivalent modules. Contact your software support organisation for further details.

In addition to this, you can use Forecasting in conjunction with, and to enhance, the following modules:

- Sales Analysis.
- Distribution Requirements Planning.
- Material Requirements Planning.
- Master Production Scheduling.
- Service Management.

Once Forecasting has generated forecasts, and these have been reviewed/manipulated, you can transfer into DRP and/or Material Requirements Planning/Master Production Scheduling. This can be done at Minor Item Group or Item Level.

Alternatively, this Requirements file could be downloaded to a PC for further evaluation.

Forecasting Configuration

You can operate Forecasting for a number of companies, the characteristics of each being maintained on control files. Forecasting does not use control data stored for any other module. The policies which can be controlled for each company are:

- Period (month) descriptions.
- Backup parameters.
- Forecast Formulas descriptions.
- Interface defaults.
- Sales Analysis database details.
- 12 or 13 periods forecasting.

Forecasting can take place at two levels - Minor Item Group (as defined in Inventory Management) and Item. Every group or item must have a profile record in this module.

The Item code used in Forecasting must match the one in Inventory Management. If sales history is required then it is extracted from Sales Analysis files.

Forecasting enables the creation and transfer of forecasts for individual locations or groups of locations. This is achieved by defining Forecasting Models.

A Forecasting Model is any logical grouping of Inventory stockrooms. This may represent a Central Warehouse, a Distribution Centre, a Manufacturing facility, or any other meaningful association.

You can adapt Forecasting to use either 12 or 13 periods per year. Options exist within the Forecasting module to extract items from Inventory Management and sales quantities from Sales Analysis for selected Models. Depending on whether 12 or 13 period forecasting is selected, a thirty-six or thirty-nine period sales history is extracted and stored in the module. This history is also aggregated and stored at Minor Item Group level.

Forecasts are generated and maintained for selected Models.

Forecasts can be transferred into DRP and/or Production modules at Minor Item Group or Item level for selected Models.

In addition to the period spread profile in Forecasting, the spread profile can be extended to include seasonal profiles from annual to period, period to weekly, and weekly to daily when transferring forecasts into DRP and/or Production modules.

Backup routines are built into the Forecast Period-End, and may be selected, with relevant parameters on the Company Profile.

Forecasting Methods

Seven forecasting methods are available within this module. These methods vary slightly according to whether 12 or 13 periods per year have been selected. Below is a list of abbreviations used in the forecast analysis:

- NPF - Next Period Forecast.
- L3P - Sum of last 3 periods sales.
- L12P/L13P - Sum of last 12 or 13 periods sales respectively.
- 2NPLY - Two times next period sales last year.
- NPYBL - Next period sales year before last.
- N3PLY - Sum of next 3 period sales last year.
- L3PLY - Sum of last 3 period sales last year.
- LP-5 - Sales 6 periods ago.
- LP-4 - Sales 5 periods ago.
- LP-3 - Sales 4 periods ago.
- LP-2 - Sales 3 periods ago.
- LP-1 - Sales 2 periods ago.
- LP - Sales last period.

Each method corresponds to a separate computer program. Addition of further forecasting methods require new programs to be written. Contact your software support organisation for further details. However, the description held against any of the existing seven forecasting methods may be amended on the Company Profile. The following descriptions relate to 12 periods a year.

Forecast Method A

Formula: $NPF = L3P / 3$

Method A calculates the next period forecast as the sum of the last 3 period sales divided by 3.

For example: If the Item had the following sales history for the last 36 periods, and method A was chosen, the forecast would be as shown:

Period	01	02	03	04	05	06	07	08	09	10	11	12
3 Years ago	402	912	704	588	762	581	728	1004	698	412	802	879
2 Years ago	1259	853	695	690	939	655	661	697	880	528	861	1096
Current year	1161	1137	926	820	841	761	752	961	759	702	733	818
Forecast	751	767	779	766	771	772	770	771	771	771	771	771



Note: The first forecast is a third of periods 10, 11 and 12 and the second forecast is a third of periods 11, 12 and 01.

From the above example,

NPF for Period 1 = $(702 + 733 + 818)/3 = 751$

Forecast Method B

Formula: $NPF = L12P/12$

Method B calculates the next period forecast as the sum of the last 12 period sales divided by 12.

For example: If the Item had the following sales history for the last 36 periods, and method B was chosen, the forecast would be calculated as shown:

Period	01	02	03	04	05	06	07	08	09	10	11	12
3 Years ago	402	912	704	588	762	581	728	1004	698	412	802	879
2 Years ago	1259	853	695	690	939	655	661	697	880	528	861	1096
Current year	1161	1137	926	820	841	761	752	961	759	702	733	818
Forecast	864	840	815	806	804	801	805	809	796	799	808	814



Note: The first forecast is a twelfth of periods 1 to 12, and the second forecast is a twelfth of periods 2 to 12 and forecast 01.

NPF = $(1161 + 1137 + 926 + 820 + 841 + 761 + 752 +$

for $961 + 759 + 702 + 733 + 818)/12$
Period
1 $= 1037/12 = 864.25$



Note: If 13 periods have been specified, then the formula is based on L13P/13.

Forecast Method C

Formula: $NPF = (2NPLY + NPYBL)/FACTOR$

Method C calculates the next period forecast as twice next period sales last year, plus next period sales year before last, divided by a factor.

To determine factor, calculate $NPLY/NPYBL$.

If this is greater than or equal to 1.5 use 2 as the factor.

If this is less than or equal to 0.25 use 4 as the factor.

If the result falls between 0.25 and 1.5 use 3 as the factor.

For example, if the Item had the following sales history for the last 36 periods and formula C was chosen, the forecast would be calculated as shown:

Period	01	02	03	04	05	06	07	08	09	10	11	12
3 Years ago	402	912	704	588	762	581	728	1004	698	412	802	879
2 Years ago	1259	853	695	690	939	655	661	697	880	528	861	1096
Current year	1161	1137	926	820	841	761	752	961	759	702	733	818

Forecast	1194	1042	849	777	874	726	722	873	799	644	776	911
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Note: The forecast for period 01 is calculated as follows:

NPLY = 1161
NPYBL = 1259
 $NPLY/NPYBL = 1161/1259$
NPF = 0.922 therefore the factor will be 3
(that is, value between 0.25 and 1.5)
 $= ((2 \times 1161) + 1259)/3 = 1193.66$

Forecast Method D

Formula: $NPF = (2NPLY + NPYBL)/3$

Method D calculates the next period forecast as twice next period sales last year plus next period sales year before last divided by 3.

For example, if the item had the following sales history for the last 36 periods and method D was chosen, the forecast would be calculated as shown:

Period	01	02	03	04	05	06	07	08	09	10	11	12
3 Years ago	402	912	704	588	762	581	728	1004	698	412	802	879
2 Years ago	1259	853	695	690	939	655	661	697	880	528	861	1096
Current year	1161	1137	926	820	841	761	752	961	759	702	733	818

Forecast	1194	1042	849	777	874	726	722	873	799	644	776	911
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Note: The forecast for period 01 is calculated as follows:

NPLY = 1161

NPYBL = 1259

NPF = $((2 \times 1161) + 1259) / 3 = 1193.66$

Forecast Method E

Formula: $NPF = N3PLY / 3$

Method E calculates the next period forecast as the sum of the next 3 period sales of last year divided by 3.

For example, if the item had the following sales history for the last 36 periods, and method E was chosen, the forecast would be calculated as shown:

Period	01	02	03	04	05	06	07	08	09	10	11	12
3 Years ago	402	912	704	588	762	581	728	1004	698	412	802	879
2 Years ago	1259	853	695	690	939	655	661	697	880	528	861	1096
Current year	1161	1137	926	820	841	761	752	961	759	702	733	818
Forecast	1075	961	862	807	785	825	824	807	731	751	875	951

The forecast for period 01 is calculated as follows:

$$\text{NPF} = (1161 + 1137 + 926)/3 = 1074.66$$

Forecast Method F

Formula: $\text{NPF} = ((\text{N3PLY}/\text{L3PLY}) \times \text{L3P}) / 3$

Method F calculates the next period forecast as the sum of the next 3 period sales last year divided by the sum of the last 3 period sales last year, times the sum of the last 3 period sales divided by 3.

For example, if the item had the following sales history for the last 36 periods, and method F was chosen, the forecast would be calculated as shown:

Period	01	02	03	04	05	06	07	08	09	10	11	12
3 Years ago	402	912	704	588	762	581	728	1004	698	412	802	879
2 Years ago	1259	853	695	690	939	655	661	697	880	528	861	1096
Current year	1161	1137	926	820	841	761	752	961	759	702	733	818
Forecast	974	778	653	602	553	576	589	589	518	515	564	624



Note: The forecast for period 01 is calculated as follows:

$$\text{N3PLY} = 1161 + 1137 + 926 = 3224$$

$$\text{L3PLY} = 528 + 861 + 1096 = 2485$$

$$\text{L3P} = 702 + 733 + 818 = 2253$$

$$\text{NPF} = ((3224/2485) \times 2253 / 3 = 974.336$$

Forecast Method G

Method G calculates the next period forecast as sales 6 periods ago times 1, plus sales 5 periods ago times 2, plus sales 4 periods ago times 3, plus sales 3 periods ago times 4, plus sales 2 periods ago times 5, plus sales last period times 6, all divided by 21 (the number of periods calculated).



Note: This formula equates to a simple single-level exponential smoothing approach with a smoothing factor of around 0.15.

For example, if this item had the following sales history for the last 36 periods and formula G was chosen, the forecast would be calculated as shown:

Period	01	02	03	04	05	06	07	08	09	10	11	12
3 Years ago	402	912	704	588	762	581	728	1004	698	412	802	879
2 Years ago	1259	853	695	690	939	655	661	697	880	528	861	1096
Current year	1161	1137	926	820	841	761	752	961	759	702	733	818
Forecast	778	775	770	773	776	776	775	775	775	775	775	775

The forecast for period 01 is calculated as follows:

$$\begin{aligned} \text{NPF} &= [(752 \times 1) + (961 \times 2) + (759 \times 3) + \\ &\quad (702 \times 4) + (733 \times 5) + (818 \times 6)] / 21 \\ &= 16332 / 21 = 777.71 \end{aligned}$$

Maintenance of Reference Data

This module uses minimal primary data from the Inventory Management module. You must, however, enter additional information to obtain maximum benefit from the module, such as:

- A generic search on Item code.
- Minimum use of codes and where they are employed a prompt is always available to assist in selecting the correct one.

Selection is normally from a displayed list. Information for the following may be maintained:

- Models.
- Group/Item Profiles.
- Demand History.
- Spread Profiles.

Forecasting Models

You can create and transfer forecasts for individual (or groups of) locations using Forecasting Models. For each Forecasting Model, the following details are maintained:

- Model Code.
- Model Description.
- Model Type.
- Target Model - Any DRP Distribution Centre, or Production Model, dependent on the entered Model Type.
- Simulation Period.
- Stockrooms - Any valid Inventory-defined stockroom directly attached to the entered DRP DC/Centre or Production Model.

Group/Item Profiles

To process a Model, a profile must be maintained for each Item or Group of Items in the Model. Group/Item profiles contain details including:

- **Model Code** - assigns the Item or Group of Items to a Forecast Model. Items and Groups of Items can be assigned to several Models. This enables forecasts created from the same Sales Analysis data to be maintained and transferred separately.
- **Simulation Period** - the number of months the month end run simulates when selecting the most accurate forecast method.
- **Locked in Forecast Method** - a specific forecast method can be locked in for the group or item. The lock in forecast will be used to calculate the next 12 or 13 periods respectively, (no simulation is performed). Alternatively, the item can be flagged as manually forecasted. In this case the Period-end will ignore the item. If no locked in or manual method is specified, all forecast methods will be evaluated.
- **Conversion Factor** - this is an equivalent units conversion factor which allows you to convert items sold in different pack sizes into equivalent units. The equivalent units are accumulated to Item group level on all reports that print forecast quantity.
- **Marketing Masks** - these direct the program to adapt the calculated forecast for a selected group of items or an individual item so as to correspond to a changing market situation which is not evident in the past. They are particularly useful for promotional or new items. The marketing masks consist of 12 or 13 sets of adjustment factors which correspond to the next 12 or 13 periods. These factors are multiplied by the original forecast.

Demand History

For each group of items, or individual item, the last three years' demand history may be amended for selected Models.

This is particularly useful for volatile items and new items just extracted from the Inventory Management module (which have no history).

Spread Profiles

Spread Profiles are required to download forecasts from this module into Distribution Requirements Planning (DRP) and Production modules. This is because these modules process forecasts on a weekly or daily basis whereas Forecasting deals with twelve or thirteen periods per year.

Processing

Processing, for each Model, can be divided into three distinct stages:

- Extracting Groups of Items and their sales.
- Applying Forecasting techniques.
- Downloading Forecasts.

Extraction

This involves the extraction of selected groups of items from the Inventory Management module, followed by sales history for these groups and items from Sales Analysis. You cannot create profile records until you have created the associated Model record.

You can select Items by Inventory Minor Item Group, or transfer all items together. This may be done as required. New items will be added to the Forecasting database, and existing item details will be updated.



Note: Group/Item profiles cannot be maintained in Forecasting until this combination has been extracted.

Full sales history, or last period sales only, can be selected for extraction from the Sales Analysis module. The extraction of last period sales should be done immediately after Sales Analysis Period-end has been run for the company, and before the Forecast Period-end run.

These sales quantities are extracted for every item which has a profile defined in Forecasting. Item Sales quantities are aggregated to Minor Item Group level and are also stored as part of the Model records.

The Transfer Full Sales History option will attempt to extract three years of sales demand from Sales Analysis.

Applying Forecasting Techniques

The Forecast Period-end job will attempt to calculate forecasts for the next twelve or thirteen periods respectively, for all item/item group profiles defined within Forecasting. Groups or items having a locked in forecast method immediately use the corresponding formula to create forecasts. Items with a locked in manual forecast are ignored.

Otherwise, the process of simulation is used. The module goes back in time evaluating each forecast method for each item and group of items. This period of time, or Simulation Period, is defined for each item/item group in its profile record.

The module predicts sales figures for the simulation period by applying each of the forecast methods. It then calculates the difference between the forecast sales figures and the actual sales figures for each period, that is, the error.

The Forecasting module decides which is the best method by calculating forecast errors for each method. The error is expressed in terms of a percentage.

The totals are accumulated over the Simulation Period. The absolute value of this error percentage is used to compare methods.

Simulation Example

Simulation Period=3, using 12 periods per annum. Forecast Method A=L3P/3.

Period	LP11	LP10	LP-9	LP-8	LP-7	LP-6	LP-5	LP-4	LP-3	LP-2	LP-1	LP
Actual	1162	1137	926	820	841	761	752	961	759	702	733	818

Forecast (Simulated)										824	807	731
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Forecast Error %
=

for Method A =



Note: \ | around figures indicate that the absolute value should be used in the equations, that is, minus signs are ignored.

There are seven forecasting methods used in this period end processing. For specific details of these methods refer to Forecasting Configuration.

Each of the seven methods has the Forecast error % calculated and the method with the lowest error is selected to project forecasts for the particular item or group of items.



Note: Forecasts created by the period-end run can be manually changed by using the Forecast

Manipulation option. This is identical to the Forecast Enquiry but with update facilities.



Note: Item Forecasts may be optionally aggregated up to Group Level using the Forecast Aggregation option.

Downloading Forecasts

Forecasts can be downloaded into:

- The DRP and/or Production modules. This may be done at Minor Item Group or Item Level.
- A specified DRP Centre DC or a Production model. This will apply the DRP/Production Spread Profiles set up within Forecasting Maintenance and distribute the forecasts over working weeks/days.
- A requirements file. This may then be removed from this module for further evaluation and downloaded into your own modules. Software programs must be written to achieve this.

Enquiries

You can select an enquiry to see Forecasts for a specific item, or a group of items within a selected model. (A model is a grouping of Inventory stockrooms).



Note: Forecasts for the group are generated by the Period-end run using Sales History quantities stored for the group. Forecasting treats the group as just another set of figures to be evaluated.

The Forecast Enquiry displays the last 3 years sales demand figures, together with the forecasts for the next year.

Also shown is the code for the forecast method selected together with the calculated forecast error.

The forecast method code can be changed to one of the other seven. This results in the forecast quantities being recalculated and displayed. No update is performed, as this is for viewing the impact of changes without affecting the data.

A Bar Chart is also available for displaying these quantities pictorially.

Reports

As part of the general policy of modules, selection parameters are available where appropriate before a report is run.

In general, reports are printed on standard listing paper with 132 print positions and 6 lines to the inch.

All reports are printed with the procedure or program number in the top left corner and the page number, date and time in the top right.

The reports can be grouped into one of three categories:

Forecast Details

The Demand Forecast report provides a detailed listing of all forecasting items, showing the forecasts for the next 12 or 13 periods. Cost and/or sales values may also be output on the report. A range of models and groups, or all items, may be selected. Totals are accumulated and printed for groups.

The Detail Forecast report shows the last three years' sales history as well as the forecasts for the next year. This is by quantity only. Selection and totals are as per the Demand Forecast report.

Forecast Performance

The Forecast History report outputs details of historical forecast, and the actual sales demand for comparison. This shows only the latest forecast for each period, together with the actual sales.

The Forecast Evaluation report shows historical and predicted forecasts for each period, together with the actual sales for comparison. Up to 6 periods of historical and 12/13 periods of predicted forecasts are included.

Static Information

Item profiles, Group profiles and Marketing Mask details may be selected for printing.

Implementing Forecasting

After installing the software provided, the recommended steps for implementing Forecasting are outlined as follows:

1. Define and create the Company Profile
2. Define and create Forecasting Models
3. Run the Transfer Item Details processing option.
4. If necessary, review the Profile records created by Step 3, particularly those allocated to the dummy Group 999, as these have no Minor Item Group defined to them in the Inventory Items file.
5. Re-run the Transfer Item Details option, if necessary.
6. Run the Transfer Full Sales History processing option.
7. If necessary, review/maintain the demand history created by Step 6 via the appropriate maintenance option, or the Detail Forecasts report.
8. At the end of the current period after Sales Analysis period-end has been run, the Forecasting Transfer Last Period Sales processing option must be run.
9. If necessary, review/maintain the last period's demand quantity created by Step 8 via the appropriate maintenance option.
10. Run the Forecast period-end processing option.
11. Review/maintain the forecasts created by Step 10.



Note: Step 8-11 must then be followed for subsequent periods & can be scheduled to run automatically via Machine Manager

Preparing to Use Forecasting

Preparing to use the Forecasting module requires careful planning and preparation. Your existing Forecasting procedures and data need to be transferred to the module.

Entering Basic Data

You must initialise your module by entering the following basic data:

- **Company Profile** - This contains control and parameter information operating your company, it also defines any links to attached modules.

The primary information maintained per company is:

- Company name.
- Current period and year.
- Period names.
- Backup parameters.
- Activate Conversion Flags (Yes/No).
- Forecast History Save Period (in periods).
- Forecast method descriptions.
- Seasonal profiles flag.
- Interface Company Code.
- Sales Analysis database details.
- Default Forecasting model (optional).

Reference Data

Within each company you should also maintain the following Reference Data:

- Forecasting Models - These models enable the creation and transfer of forecasts for individual locations or groups of locations.
- Items - These must be defined in the Inventory Management Items file. Forecasting extracts items from this source to create Profile records in the Forecasting database.
- Minor Item Groups - These should be defined in the Inventory Management Parameters file and allocated to appropriate items in the same module. It must also be defined in the Sales Analysis module for the appropriate company.

Forecasting extracts:

- The selected Groups from Inventory and creates a Profile record for each Group within the Forecasting database.
- Invoiced sales from Sales Analysis at item level, and accumulates these quantities at Minor Item level before updating them in Forecasting at both levels.

The Forecasting module expects that the stockroom (depot) field, Minor Item Group and quantity fields exist within the Sales Analysis database. If this is not the case, then please contact your software support organisation.

- Demand History - The demand history for the last three years is created using the Transfer Full Sales History processing run.
- DRP Spread Profile - If Distribution Requirements Planning is installed, then a Spread Profile is used to split the period forecast quantities into weekly or daily quantities for DRP.

You have the option of specifying a weekly Spread Profile in Forecasting, or to use the weekly and daily Spread Profiles defined in DRP. You do not need to set up both - only one will be used (based on the Company profile Seasonal Profiles setting.)

When defining the Spread Profiles, Forecasting displays the appropriate DRP weekly calendar for each period. Against each week a percentage is entered.

The second option automatically recognises the spread profiles already defined and maintained in DRP. These Spread Profile records are used by the Transfer Forecasts to DRP option within Forecasting.

- MPS/MRP Spread Profile - If Master Production Scheduling and/or Material Requirements Planning are installed, Spread Profiles can be used for the same purpose as described previously.